Proposal for a Data Science Concentration in Business Analytics at Carthage College

Introduction

In an increasingly data-driven business world, the ability to find meaningful insights with data is so important. With my 3 course concentration combined with the data science courses, I hope to create a good foundation of data science knowledge with a business administration focus. I will use the combination of courses to give me a good understanding of how to use data science to interpret data to make valuable business decisions. I will be able to identify trends and optimize business procedures to make a real impact on the business world.

The Data Science Major consists of the following 11 core courses (38 cr.).

- CSC 1030 Data Science I (4 credits)
- CSC 1040 Data Science II (4 credits)
- MTH 1120 Calculus I (4 credits)
- MTH 2040 Linear Algebra (4 credits)
- MTH 3090 Mathematics of Data Science (4 credits)
- MTH 3350 Data Science Portfolio I (1 credit)
- MTH 4350 Data Science Portfolio II (1 credit)
- MTH 1050 Elementary Statistics (4 credits)
- CSC 1810 Principles of Computer Science I (4 credits)
- CSC 2810 Database Design and Management (4 credits)
- MTH 2190 Ethics in Data Science (4 credits)

My Concentration

As part of the Data Science major, students are required to complete an applied concentration to demonstrate how data science techniques can be integrated into a specific domain. The purpose of this concentration is to help students contextualize their technical skills in real-world applications and better prepare for careers in specialized fields. To fulfill this requirement, I propose a concentration in Business Analytics by adding the following three courses:

- **MGT 3100 : Introduction to Business Analytics**
    - This course is described as a survey of the mathematical models of Management Science and Operations Research (such as linear programming, queuing theory, decision analysis, and simulation) applied to managerial decision making (Carthage College 2024-2025 Catalog, 96).
        - *MGT 3100: Introduction to Business Analytics* will serve as the foundation of my concentration by introducing key analytical tools used in business decision-making. The course will expose me to fundamental models and techniques that are central to solving real-world problems using data. These initial concepts will guide my understanding of how businesses can use analytics to improve efficiency, make strategic choices, and optimize operations.
- **MKT 3130: Marketing Principles**
    - This course provides a general knowledge of marketing, emphasizing marketing mix elements (Product, Price, Promotion, and Placement) for both consumer and industrial products, marketing strategies, customer behavior, and promotion. Topics include situation analysis techniques, marketing segmentation, identification of target markets, product/brand positioning via the 4 Ps of marketing, development and presentation of a marketing plan for a local

business, and the social, ethical, and legal issues in marketing (Carthage College 2024–2025 Catalog, p. 98).

- *MKT 3130: Marketing Principles* will provide me with a foundational understanding of marketing strategies and consumer behavior, an important part in understanding business analytics. This background will be valuable when interpreting data related to customer trends and designing marketing strategies.

- **ACC 2000: Accounting Fundamentals**
  - This course is described as a combination of relevant portions of ACC 2010 Financial Accounting and ACC 2020 Managerial Accounting, to provide a comprehensive introduction to accounting concepts, financial statements, and the usage of accounting information for decision making. The emphasis is on the application of financial data in the business world for planning, performance evaluation, and tactical decision-making. Topics include financial ratio analysis, evaluation of capital investments, cost-volume-profit analysis, inventory valuation, and make vs. buy analysis (Carthage College 2024-2025 Catalog, 24).
    - *ACC 2000: Accounting Fundamentals* will round out my concentration by providing a solid grasp of how financial data is structured, interpreted, and used in business decisions. This course will help me connect the analytical techniques learned in MGT 3100 with financial strategies and business operations. By understanding accounting data, I will be able to develop more accurate, data-informed models.

Conclusion

This three-course concentration in Business Analytics will allow me to apply data science principles directly to business decision-making. It will give me the skills to analyze complex data sets and provide actionable insights for businesses. By completing the core Data Science curriculum along with these applied business courses, I will develop a strong foundation in business-focused analytical techniques.

After specializing in this, I will be well-prepared for job roles such as Business Data Analyst, Data Analyst, Business Intelligence Analyst, Operations Analyst, or Financial Analyst. In my career, I hope to work in a role where I can help organizations make better decisions using data. I will be prepared to work in diverse sectors such as retail, finance, healthcare, or technology. I plan to achieve this by using data to understand customer behavior, improve business operations, guide financial decisions, and strengthen their marketing. Ultimately, this concentration will position me to succeed in the evolving field of Business Analytics and to make a meaningful impact on the business world through data-driven solutions.

References

Carthage College, Carthage College 2024-2025 Catalog, 2025.